

high. Let us suppose, when switch S1 is pressed, I1 input of IC1 goes high and its corresponding output Q1 goes high. As a result, current passes through D5 to piezo buzzer PZ1, which creates a beep. At the same time, current also passes through diodes D6-D7 to show

the number on the LED display.

Similarly, when any other switch (S2-S4) is pressed, the corresponding number gets displayed on 7-segment display DIS1 and buzzer sounds. Switch S5 is used to reset the display exclusively. The circuit is powered by 9V battery.

Assemble the circuit on a general-purpose PCB and enclose it in a suitable case along with the 7-segment display and piezo buzzer. The assembled circuit can be kept near the host and the switches connected externally can be assigned to the players. ●

MONOLITHIC STEP-DOWN SWITCHING REGULATOR

■ S.N. BANERJEE

L4962 is a monolithic step-down switching regulator. It provides output voltage of 5.1V to maximum 40V, delivering 1.2A to 2.5A current, depending on the type and package of the IC. Their theoretical internal functions are almost the same. The heart of the device is the regulation loop consisting of a

zener zap trimmed to ± 2 per cent). This error signal is then compared with the sawtooth signal to generate the fixed frequency pulse width modulated pulses, which drive the output stage.

Fig. 1 shows circuit diagram of the regulator. The gain and frequency of the loop can be set by RC network connected to IC pin 11. When the loop is closed directly by connecting the

supply output to the feedback input IC pin 10, an output voltage of 5.1V is produced. Higher output voltages are obtained by inserting a voltage divider in this feedback path.

Output over-currents at switch-on are prevented by the self-start function. The error amplifier output is initially clamped by the external capacitor C_{ss} at pin 15, and is allowed to rise linearly as this capacitor is charged by a constant cur-

rent source.

Output overload protection is provided in the form of a current limiter. When the load current exceeds a preset threshold, this comparator sets a flip-flop, which disables the output stage and discharges the soft start capacitor.

Another internal comparator resets the flip-flop when the voltage across capacitor C3 falls to 0.4V. The output is thus re-enabled and the voltage rises under the control of soft start network. If overload condition is still present, the limiter triggers again when the threshold current is reached. The average short-circuit is limited to a safe value by the dead time introduced in the soft start network. The thermal overload circuit disables circuit operation when the junction temperature is about 150°C and has hysteresis to prevent instability. Frequency is about 100 kHz with parallel RC network connected to this terminal.

Assemble the circuit on a gen-

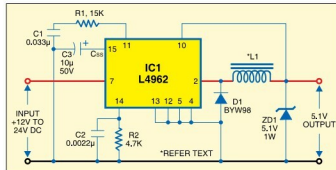


Fig. 1: Circuit of switching regulator

sawtooth oscillator, error amplifier, comparator and source-sink output stage.

An error signal is produced by comparing the output voltage with a precise 5.1V on chip reference (which



Fig. 2: Specification of L1

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